

Abrasive Reference Chart | Metalsmithing

Covers sheet/paper abrasives and cloth abrasives relevant to bench metalsmithing and jewelry work.

SHEET / PAPER ABRASIVES

	Silicon Carbide	Aluminum Oxide	Ceramic / Zirconia Alumina	Diamond (Sheet/Film)	Garnet
Mohs	~9.5	~9	~9	10	6.5–7.5
Wear Behavior	Friable — fractures to expose fresh cutting edges	Dulls progressively — no fracturing	Pressure-induced micro-fracture; controlled self-sharpening	Bonding medium fails before abrasive wears	Dulls and fractures inconsistently
Durability	Moderate	Good	Very good	Exceptional	Poor
Clog Resistance	Good (especially wet)	Moderate	Good	Very good	Poor
Scratch Consistency	Excellent	Good when fresh, degrades over time	Very good	Excellent	Inconsistent
Backing Options	Paper, Cloth, Film, Foam	Paper, Cloth, Film	Cloth, Belt, Disc	Film, Sheet, Stick	Paper
Wet Use	Yes	Sometimes	Varies	Yes	No
Best For	General metal finishing; nonferrous metals; fine finishing sequences	Ferrous metals; rough stock removal; hard alloys	Rapid stock removal; steel; titanium; power tool use	Tungsten, carbide; ultra-fine finishing; lapping	Wood only — avoid for metalsmithing

CLOTH ABRASIVES

	Emery Cloth	Silicon Carbide Cloth	Aluminum Oxide Cloth
Mohs	7–9 (variable)	~9.5	~9
Wear Behavior	Dulls with no self-sharpening; inconsistent due to natural mineral composition	Friable — same as SiC paper but on durable backing	Dulls progressively
Durability	Moderate (backing outlasts abrasive)	Very good	Good
Clog Resistance	Poor–Moderate	Good	Moderate
Scratch Consistency	Inconsistent — natural mineral with variable grit sizing	Excellent	Good when fresh
Backing Options	Cloth only	Cloth only	Cloth only
Wet Use	Tolerates it	Yes	Sometimes
Best For	Ring shanks; mandrel wrapping; inside bezels — tasks requiring flexible, tear-resistant backing	Same tasks as emery cloth with superior cut and longevity — straightforward upgrade	Ferrous metals where cloth flexibility is needed; common in roll/belt form

Key notes: Silicon carbide cloth is a direct upgrade over emery cloth in every performance category. Emery cloth persists on the bench due to availability and habit, not merit.

CAMI vs. FEPA grit scales: Two competing grading standards produce numbers that look similar but are not interchangeable at fine grits. CAMI (US standard) uses mesh screen sizing. FEPA (European standard, marked with a P prefix — P220, P400, etc.) uses laser diffraction particle sizing, which is more precise. The systems track closely at coarse grits but diverge significantly at fine grits: CAMI 400 ≈ FEPA P800, and CAMI 600 ≈ FEPA P1200. Mixing systems within a finishing sequence can leave unintended gaps or redundancies in scratch refinement. Most professional and imported abrasives use FEPA; most US commodity hardware-store product uses CAMI, often without stating which system.